PROJECT REPORT

DataPulse: Superstore Data Analytics & Sales Prediction Dashboard

Using Python, Streamlit, Excel, SQLite & Machine Learning

1. ABSTRACT

DataPulse is an intelligent Superstore Data Analytics and Sales Prediction Dashboard developed using Python, Streamlit, SQLite, Pandas, and Machine Learning. The system allows users to securely log in, upload CSV/Excel datasets, perform data quality analysis, clean and filter data, visualize insights through interactive charts, and store processed data in an SQLite database. A secure database download feature with username-password authentication is integrated for protected access. The system generates future sales forecasts, prediction graphs, and high/low sales classification, helping businesses make data-driven decisions.

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2. AIM

To develop a secure interactive business intelligence dashboard that performs:

- Superstore data analytics
- Data cleaning
- Visualization
- SQLite database management
- Machine learning-based sales forecasting

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3. OBJECTIVES

- Upload and analyze Superstore datasets
- Perform null value handling
- Detect and remove duplicate records
- Enable dynamic data filtering
- Generate KPI business metrics
- Create visual analytics charts
- Store cleaned data in SQLite
- Securely download SQLite database
- Predict future sales using ML
- Classify production as High/Low

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4. TOOLS AND TECHNOLOGIES USED

Component	Technology / Libraries
Programming Language	Python
Framework	Streamlit
Database	SQLite
Data Analysis	Pandas
Numerical Computing	Numpy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn
IDE	PyCharm
File Support	CSV, Excel

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6. SYSTEM ARCHITECTURE

User Login
↓
Upload CSV / Excel
↓
Data Quality Check
↓
Data Cleaning
↓
Data Filtering
↓
KPI Metrics
↓
Visualization
↓
SQLite Storage
↓
Secure Database Download
↓
Machine Learning Prediction
↓
Sales Forecast Graph & Table

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7. DATABASE DESIGN

Column name	Description
Order_Date	Order date
Ship_Mode	Shipping type
Region	Sales region
Category	Product category
Sub_Category	Product subcategory
Sales	Sales amount
Quantity	Quantity sold
Profit	Profit amount
Sales_Level	High/Low classification

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8. MODULES IMPLEMENTED

8.1 User Authentication Module
Features:

- Username login
- Session-based authentication
- Logout functionality

8.2 File Upload Module

Supports:

- CSV files
- Excel (.xlsx) files

Functions:

- Read uploaded dataset
- Display raw data

8.3 Data Quality Check Module

Checks:

- Number of rows
- Number of columns
- Duplicate records
- Null values

8.4 Dynamic Filtering Module

Available Filters:

- Ship Mode
- Region
- Category
- Sub Category

8.5 KPI Metrics Module

Generated metrics:

- Total Sales
- Total Quantity
- Total Profit

8.6 SQLite Database Module

Functions:

- Save cleaned dataset
- Dynamic table creation
- Load saved tables
- View stored database records

8.7 Secure Database Download Module

Security Features:

- Username verification
- Password authentication
- Protected SQLite DB download

Allowed users:

Username	Access
Nirai	Authorized
admin	Authorized

8.8 Visualization Module

Chart	Purpose
Bar chart	Category comparison

Line chart	Trend analysis
Pie chart	Distribution analysis

8.9 Machine Learning Prediction Module

Algorithm Used: Random Forest Regressor
Input Features:

- Day
- Week
- Month
- Year
- Category
- Sub Category
- Region
- Ship Mode

Output:

- Predicted sales value
- High/Low production status

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9. MACHINE LEARNING WORKFLOW

Generate sample sales dataset
↓
Feature engineering
(Day/Week/Month/Year)
↓
Label Encoding
↓
Train-Test Split
↓
Random Forest Training
↓
Prediction
↓
Future Forecast Graph
↓
Prediction Table Download

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10. FEATURES OF THE SYSTEM

Day-wise Prediction
Predicts sales for 30 days.

Week-wise Prediction
Predicts sales for 52 weeks.

Month-wise Prediction
Predicts sales for 12 months.

Year-wise Prediction
Predicts sales up to future years.

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11. OUTPUT GENERATED

The system generates:

- Raw Data Table
- Cleaned Data Table
- KPI Metrics
- SQLite Database Tables
- Secure Download Button
- Predicted Sales
- High/Low Status
- Forecast Graph
- Prediction Table
- CSV Download

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12. PROBLEMS FACED AND SOLUTIONS

Problem	Solution
Null values	Mean/Median/Mode
Duplicate records	Duplicate removal
Streamlit errors	Debugged code
Name error issues	Added default variables
SQLite integration	Used sqlite3
Category encoding	LabelEncoder
Prediction inconsistency	Trained Random Forest properly
Graph issues	Modified plotting logic

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13. ADVANTAGES

- Easy to use
- Interactive dashboard
- Secure database access
- Supports multiple file formats
- Automated data cleaning
- Business insights generation
- Future sales forecasting
- Downloadable predictions
- Lightweight SQLite storage

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14. LIMITATIONS

- Depends on data quality
- Requires correct column names
- Basic login security
- Sample generated prediction data
- Large datasets may slow performance

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15. CONCLUSION

The DataPulse project successfully integrates data analytics, database management, visualization, and machine learning into a single interactive dashboard. It helps users analyze sales performance, predict future trends, and make informed business decisions efficiently.

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16. FUTURE ENHANCEMENTS

- Cloud deployment
- Real-time database connection
- Advanced authentication
- Deep learning forecasting
- Power BI integration
- PDF report generation
- Live analytics API integration

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1. Day Wise Work Memo

Daily Progress Report (April 1, 2026 – May 7, 2026)

Date	Work Completed
April 1 & April 2	Project topic finalized and Superstore dataset collected ,

	Created Streamlit project structure.
April 3 & April 4	Implemented login page using Streamlit , Added CSV and Excel file upload functionality.
April 5 & April 6	Displayed raw dataset in dashboard , Implemented data quality checking module.
April 7 & April 8	Added duplicate record detection , Implemented duplicate removal feature.
April 9 & April 10	Added null value handling using Mean method , Added Median and Mode handling methods
April 11 & April 12	Created cleaned data section , Developed dynamic filtering system.
April 13 & April 14	Added Region filter , Added Category filter.
April 15 & April 16	Added Sub Category filter , Added ship mode filter.
April 17 & April 18	Developed KPI metrics section , Added Total sales calculation.
April 19 & April 20	Added Total Quantity calculation , Added Total profit calculation.
April 21 & April 22	Implemented SQLite database connectivity , Added save to SQLite feature.
April 23 & April 24	Developed SQLite dashboard viewer , Added database download feature and username , password secure system
April 25 & April 26	Created Bar chart visualization , Created Line chart visualization.
April 27 & April 28	Created Pie chart visualization , Started Machine Learning module.

April 29 & April 30	Prepared sample prediction dataset , Added state-based features (Day, Week, Month, Year).
May 1 & May 2	Implemented Label Encoding , Built Random Forest Regressor model.
May 3 & May 4	Added prediction input system , Developed High/Low classification system.
May 5, 6 and 7	Added Day, Week, Month, Year prediction graph , Final testing, bug fixing, report preparation and project completion.

Final Result

The project successfully integrates:

- Data Analytics
- Visualization
- Database Management
- Machine Learning
- Forecasting
- Business Intelligence

into a single interactive dashboard system.

The system effectively predicts future sales and supports business decision-making through intelligent data analysis.